Animal Shelter Management System

Class and Function Explanations:

Aside from main.cpp, the system is basically split into five primary classes, each with their specific responsibilities:

A. Class: Animal

- It represents an individual animal in the shelter
- It encapsulates attributes like name, species, animalID, and isAdopted (adoption status).
- Static Member (totalAnimals): A static integer used to maintain count of all Animals currently in existence. It is incremented in the constructor and decremented in the destructor.
- Encapsulation: All attributes are private and accessed with getter/setter methods.

B. Class: ShelterUser (Abstract Base Class)

- This class is supposed to serve as a generic template for any user of the system (Whether it is a Staff or an Adopter).
- It defines common attributes name and userID.
- Abstraction: It declares a pure virtual function displayUserInfo(). This makes ShelterUser an abstract class, preventing incomplete user objects and enforcing that all derived classes implement their own display logic.
- Virtual Destructor ensures that derived class destructors are called correctly when deleting a ShelterUser pointer.

C. Class: Adopter (Derived from ShelterUser)

- Adopter represents a user who visits the shelter to adopt pets.

- Maintains a private std::vector<Animal*> (resizable array) called “adoptedPets” to track which animals this specific user has adopted.
- adoptPet(Animal*) checks if an animal is already adopted (for a possible error). If not, it updates the animal's status to true and adds it to the user's personal vector.
- returnPet(Animal*) searches the user's array for the animal. If found, it removes the animal and turns its adoption status to false.

D. Class: ShelterStaff (Derived from ShelterUser)

- It represents a staff member/shelter staff.
- addAnimal(Animal*) and removeAnimal(string) these functions simulate the authorization process. While the ShelterSystem performs the actual storage, the Staff class functions represent the action of processing these requests.

E. Class: ShelterSystem

- The main controller that manages the collections of animals and users.
- <Animal*> and <ShelterUser*> are used to store dynamic objects.
- Recursive Search**:** The searchAnimal method uses a private helper function searchHelper. This function recursively iterates through the animals array (vector) to find the desired animal by name, satisfying the recursion requirement.
- The destructor iterates through both vectors and deletes every pointer to prevent memory leaks.

Design Decisions:

The system uses ShelterUser* pointers to store both Adopter and ShelterStaff objects in a single array. This allows the system to treat all users uniformly (e.g., when calling displayUserInfo()) (Polymorphism is used here basically.).

I had to use vectors (dynamic arrays) instead of static arrays because the number of animals and users in the shelter is dynamic.

I created objects using “new” (e.g., new Animal(...)) to ensure they persist beyond a single function. This requires the ShelterSystem destructor to delete these objects and free unnecesarry memory.

Instead of a standard loop, the search functionality uses recursion. This gives the ability to for example search a list into a base case (found what was being looked for or the list has ended), and a recursive step (checks the next index).

Challenges and Solutions:

- The first thing i struggled with was using pointers in a Project with this many classes. I feel like i have a better understanding of how they work now.
- Thankfully i didn’t have to write the whole thing from scratch but i had to swap arrays(static arrays) with vectors(dynanic arrays) in the parts that arrays were being used
- When I first wrote the code without any errors in CLion, in the console i had "undefined reference" errors. This was because the CMakeLists.txt file only included main.cpp and did not link the other class’s source files (Animal.cpp, ShelterSystem.cpp, etc.). I listed every .cpp class and fixed it.
- The total animal count in the output was displaying "2" immediately because I instantiated the test objects before adding to the system. I adjusted main.cpp to instantiate and add animals in the correct order.

Sample Outputs:

I forgot to take the screenshot of the first output i got but it was:

```
g++.exe "C:\Users\User\Documents\C++ Projects\first\main.cpp" -o main

C:\Program Files\JetBrains\CLion 2025.2.4\bin\mingw\bin/ld.exe:
C:\Users\User\AppData\Local\Temp\ccGHaJbK.o:main.cpp:(.text+0x4b):
undefined reference to `ShelterSystem::ShelterSystem()'

C:\Program Files\JetBrains\CLion 2025.2.4\bin\mingw\bin/ld.exe:
C:\Users\User\AppData\Local\Temp\ccGHaJbK.o:main.cpp:(.text+0x99):
undefined reference to `Adopter::Adopter(std::__cxx11::basic_string<char,
std::char_traits<char>, std::allocator<char> >, int)’...

Process finished with exit code 1
```

If i wrote it all it would be way too long but esentially this is what made me change “CMakeLists.txt”.

```
Animal Shelter System Initialized.

Adding new adopter: John Doe (ID: 101)
Adding new staff: Sarah Smith (ID: 102)
Staff Sarah Smith is processing addition of: Buddy
Adding animal: "Buddy" (Dog), ID: D101
Total animals: 2

Staff Sarah Smith is processing addition of: Whiskers
Adding animal: "Whiskers" (Cat), ID: C102
Total animals: 2

Adopter John Doe is adopting "Buddy".
Adopter John Doe returned "Buddy".

Displaying all animals:
Name: Buddy, Species: Dog, ID: D101, Adopted: No
Name: Whiskers, Species: Cat, ID: C102, Adopted: No
Total animals: 2

Displaying all users:
Adopter: John Doe, ID: 101, Adopted Pets: 0
Staff: Sarah Smith, ID: 102

[Test] Recursive Search for 'Whiskers':
Found: Name: Whiskers, Species: Cat, ID: C102, Adopted: No

Process finished with exit code 0
```

The obvious major issue here was that the total animals counter immediately jumped to two.

```
Animal Shelter System Initialized.

Adding new adopter: John Doe (ID: 101)
Adding new staff: Sarah Smith (ID: 102)

Adding animal: "Buddy" (Dog), ID: D101
Total animals: 1

Adding animal: "Whiskers" (Cat), ID: C102
Total animals: 2

Adopter John Doe is adopting "Buddy".
Adopter John Doe returned "Buddy".

Displaying all animals:
Name: Buddy, Species: Dog, ID: D101, Adopted: No
Name: Whiskers, Species: Cat, ID: C102, Adopted: No
Total animals: 2

Displaying all users:
Adopter: John Doe, ID: 101
Staff: Sarah Smith, ID: 102

Process finished with exit code 0
```

And this is the corrected final version.